Topic Modeling, Sentiment and Election Markets

Applied LSA and sentiment analysis around the 2026 US midterms

FRE-GY 7871 · Assignment 5 · sample 2026-03-02 to 2026-10-09 (155 NYSE sessions) · 5,816 filtered tweets · built 2026-10-10T13:22

Latest consensus P(Dem): House 90.3%, Senate 61.8%, Democratic sweep 61.0%.

Executive summary

Stage 1 (sentiment → election odds). 0 of 21 tests survive the Benjamini–Hochberg correction at 10% (smallest q = 0.56); 1 of 21 reject at an unadjusted 5% level, about what chance alone delivers. At a daily horizon, public issue sentiment on X does not measurably lead prediction-market prices.

Covariance SVD. No factor of the lagged sentiment–return cross-covariance beats its circular-shift permutation null (p = 0.73, 0.24, 0.54); the leading factor's out-of-sample correlation is -0.29 (t = -2.3), the opposite sign to its training fit, which is what over-fitting looks like.

News check (GDELT). 0 of 48 tests of per-issue news salience and news tone on ΔProb survive FDR at 10% (smallest q = 0.11). X discourse tracks the news agenda most closely for Housing Affordability (ρ = 0.53), Foreign Policy/Defense (ρ = 0.50), Healthcare (ρ = 0.49); the median level correlation across issues is 0.31.

Stage 2 (election odds → assets). 7 of 57 excess-return tests and 0 of 57 volatility tests survive FDR at 10%.

Robustness. 5 of 7 survivors pass every check (HAC errors; without the 5 largest ΔProb sessions; without the 5 Jun 2026 sell-off; same sign in both halves; null placebo): ΔP(Sweep) → XLF (Σγ = -0.22, p = 0.003, both halves); ΔP(Sweep) → XLK (Σγ = +0.29, p = 0.002, mainly the second half); ΔP(Sweep) → UNH (Σγ = -0.46, p = 0.002, both halves); ΔP(Sweep) → NVDA (Σγ = +0.38, p = 0.006, mainly the second half); ΔP(Senate) → META (Σγ = +0.53, p = 0.004, mainly the second half). A rise in P(Dem senate) is followed the next session by outperformance of META. A rise in P(Dem sweep) is followed the next session by underperformance of XLF, UNH and outperformance of XLK, NVDA. The other survivors rest on a few extreme sessions or one half of the sample.

Rigobon–Sack. Relevance test p-values (Var_H/Var_L of ΔProb): House 0.629, Senate 0.034, Sweep 0.006. Even where the variance shift is significant it is too small to pin β down: 0 of 57 estimates have a 95% CI excluding zero, so contemporaneous causal effects are not identified in this sample.

Granger tests are predictive restrictions on lagged information, not structural effects. With one election cycle, read the q-values and the robustness battery rather than single p-values.

Data. 5,816 filtered tweets (general + affordability strata, every session covered); prices for 9 sector ETFs, 10 single names, SPY and VIX with no missing prints; prediction-market quotes on House: 155, Senate: 155, Sweep: 155 of 155 sessions; GDELT news volume and tone for every issue (19 outage days masked).

Figure 1. Implied probability of Democratic control, 16:00 ET quotes

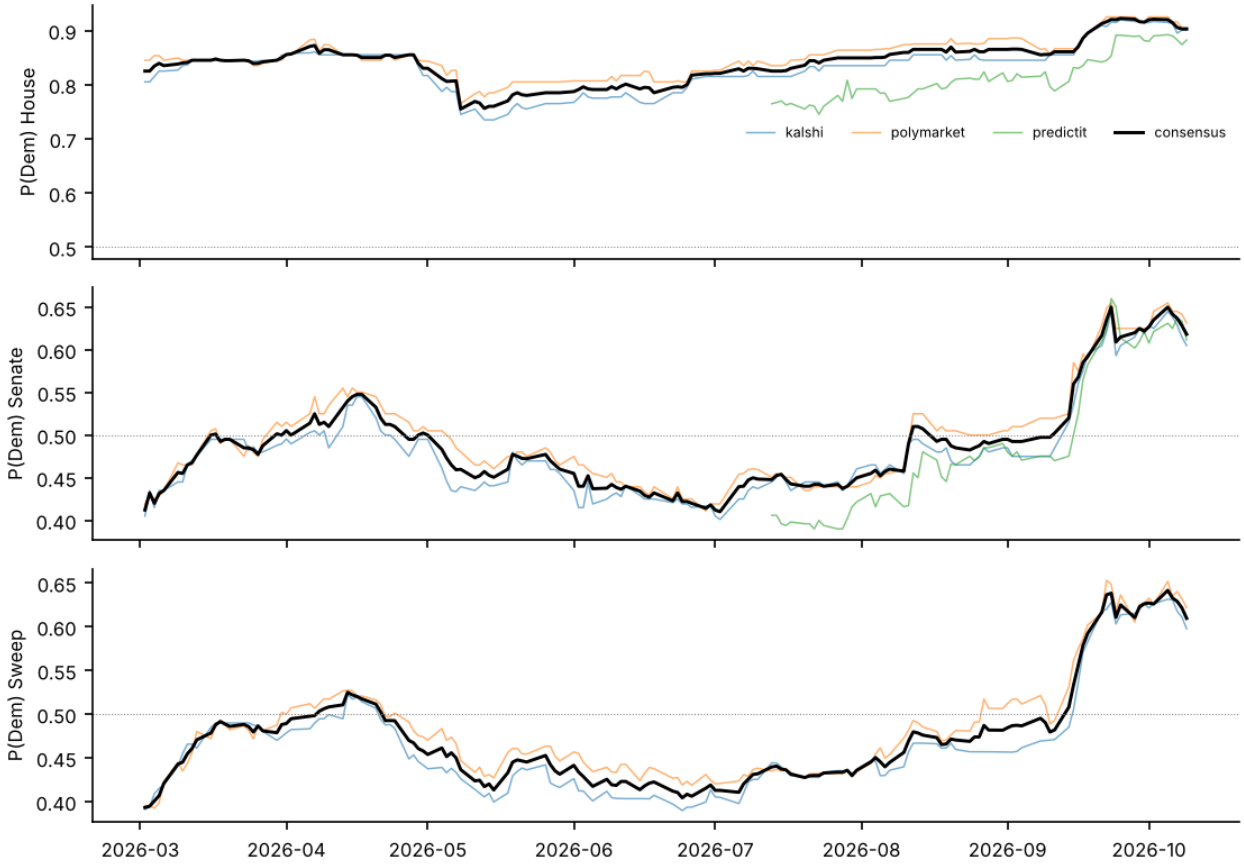
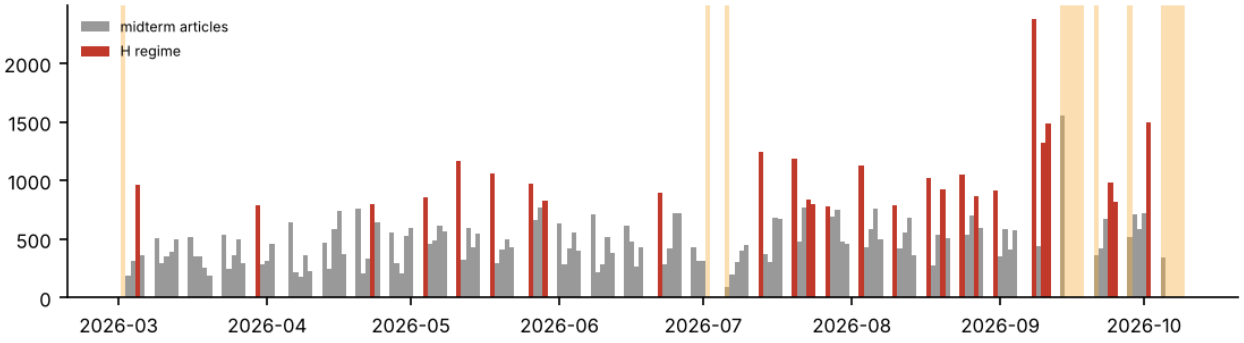


Figure 8. GDELT midterm coverage per session: H regime (red), GDELT outages (orange, excluded)



1. Topics: LSA on X discourse

Table 1A. SVD latent topics and term loadings (U_k)

Issue	Axis	σ	Seed share	Purity	Stability	Top terms (loading)
Economy/Inflation	23+	3.07	0.71	0.97	1.00	tariffs (0.73); trump's (0.17); trump's tariffs (0.13); trade (0.13); trump tariffs (0.12); canada (0.11)
Economy/Inflation	6+	4.06	0.74	0.93	1.00	gas (0.48); prices (0.39); gas prices (0.29); grocery (0.19); groceries (0.17); grocery prices (0.15)
Energy/Climate	21+	3.11	0.48	0.86	0.60	energy (0.50); climate (0.15); ballot (0.14); bills (0.13); costs (0.11); energy costs (0.10)
Foreign Policy/Defense	3+	4.63	0.79	0.94	1.00	war (0.52); iran (0.46); war iran (0.24); iran war (0.18); military (0.17); billion (0.09)
General	0+	9.79	0.07	0.36	–	trump (0.19); republicans (0.19); health (0.16); democrats (0.15); congress (0.13); people (0.13)
Healthcare	10+	3.65	0.79	0.97	1.00	care (0.59); health care (0.53); health (0.30); affordable (0.11); care costs (0.09); affordable care (0.08)
Healthcare	10-	3.65	0.23	0.71	1.00	healthcare (0.12); insurance (0.09); health insurance (0.08); house (0.07); america (0.05); housing (0.03)
Healthcare	20+	3.13	0.31	0.90	1.00	medicare (0.43); social security (0.37); security (0.37); social (0.35); security medicare (0.20); medicare medicaid (0.14)
Healthcare	5+	4.09	0.72	0.99	1.00	health insurance (0.56); insurance (0.55); health (0.45); insurance premiums (0.08); insurance companies (0.07); insurance costs (0.06)
Healthcare	8+	3.77	0.41	0.94	1.00	medicaid (0.48); cuts (0.39); cut (0.18); medicaid cuts (0.16); big (0.16); tax cuts (0.15)
Healthcare	9+	3.71	0.55	0.71	1.00	healthcare (0.55); premiums (0.30); obamacare (0.20); costs (0.19); healthcare costs (0.14); medicare (0.12)
Healthcare Affordability	12+	3.50	0.25	0.73	1.00	obamacare (0.40); money (0.29); insurance companies (0.15); companies (0.14); pay (0.14); dollars (0.13)
Healthcare Affordability	26+	2.98	0.66	0.93	1.00	aca (0.46); subsidies (0.41); aca subsidies (0.31); expire (0.17); coverage (0.15); let (0.14)
Healthcare Affordability	7+	3.88	0.68	0.65	1.00	drug (0.48); drug prices (0.42); prices (0.36); prescription (0.29); prescription drug (0.27); lower (0.16)
Housing Affordability	15+	3.38	0.50	0.93	1.00	rent (0.54); rent control (0.35); ballot (0.32); control (0.31); massachusetts (0.15); groceries (0.13)
Housing Affordability	2+	4.71	0.60	0.99	1.00	housing (0.54); housing costs (0.37); costs (0.36); lower (0.20); lower housing (0.20); bipartisan (0.12)
Immigration	17+	3.27	0.69	0.99	1.00	immigration (0.61); illegal (0.28); enforcement (0.27); immigration enforcement (0.22); illegal immigration (0.16); immigrants (0.11)
Immigration	4+	4.46	0.64	0.99	1.00	border (0.54); ice (0.31); border patrol (0.29); patrol (0.28); ice border (0.22); fund (0.14)

Varimax-rotated LSA, k = 30, uncentred TF-IDF; loadings are entries of U_k. Stability: share of 5 refits on 80% subsamples in which the pole reappears. Energy/Climate is the least stable (0.6); no Technology/AI axis forms at any k.

Figure 2. LSA spectrum and topic stability

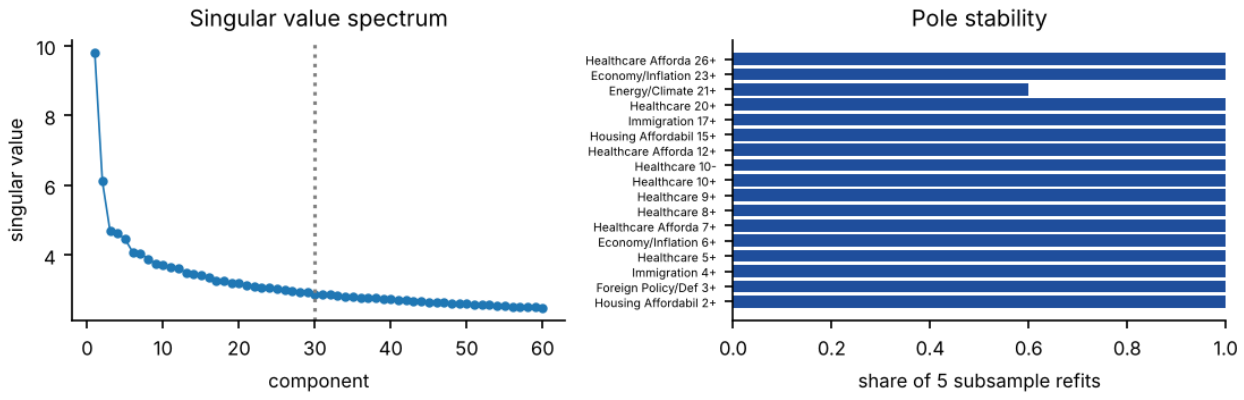


Table 1B. Topic salience and sentiment by issue

Issue	Docs	Salience	S mean	S sd	% Pos	% Neg	% Anxiety	% Anger	% Optimism	% Re-signed	Coverage	Used
Economy/Inflation	416	0.17	-0.40	0.25	25.00	71.88	14.90	6.01	14.66	13.46	0.98	yes
Housing Affordability	447	0.03	0.01	0.27	47.65	42.06	13.87	3.36	15.88	9.40	0.98	yes
Healthcare Affordability	401	0.07	-0.17	0.26	41.90	50.37	6.48	4.24	11.97	12.97	0.99	yes
Healthcare	1,076	0.27	-0.04	0.21	45.35	51.39	11.43	6.88	11.15	7.62	0.99	yes
Energy/Climate	160	0.08	0.24	0.23	72.50	26.25	13.75	3.12	31.87	5.00	0.96	yes
Immigration	480	0.25	-0.01	0.29	44.79	46.88	9.17	7.50	13.12	1.67	0.98	yes
Foreign Policy/Defense	269	0.12	-0.77	0.15	8.92	89.59	25.28	8.18	21.93	0.37	0.88	yes
Technology/AI	0	0.00	–	–	–	–	–	–	–	–	0.00	no (too sparse)

Salience: mean daily share of issue loading (general stratum only). S: 5-session pooled topic-conditioned sentiment. % Re-signed: documents whose issue-conditioned polarity differs from generic VADER.

Figure 4. Issue salience (general stratum) and topic-conditioned sentiment

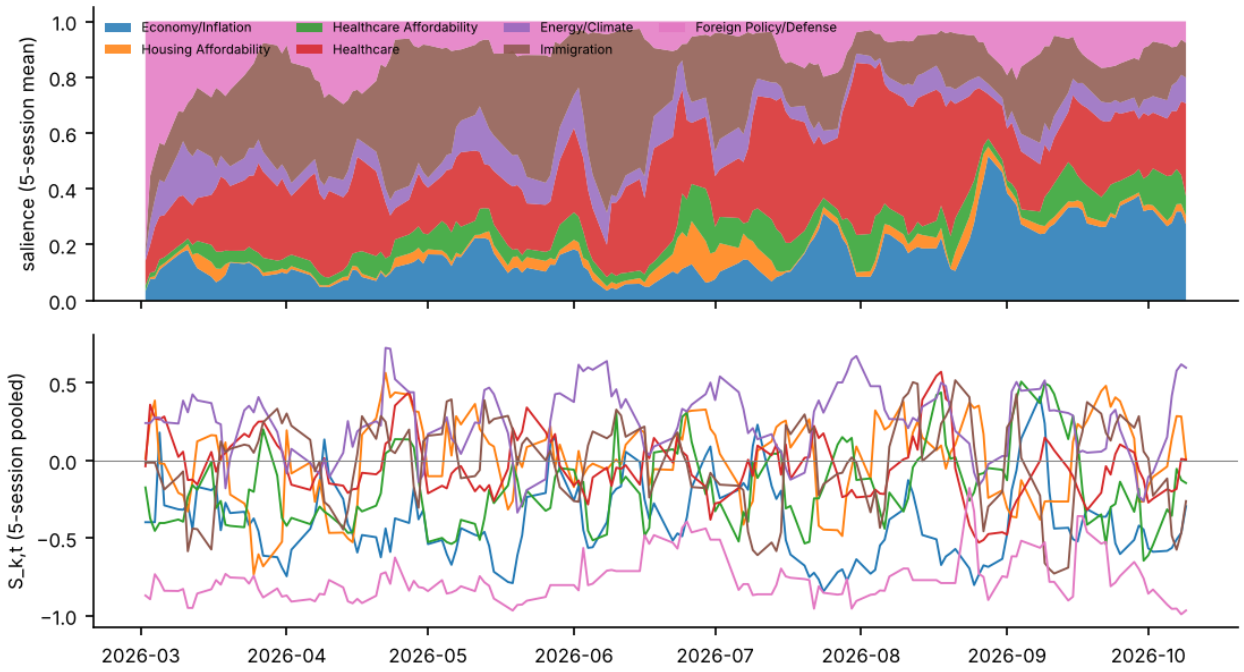
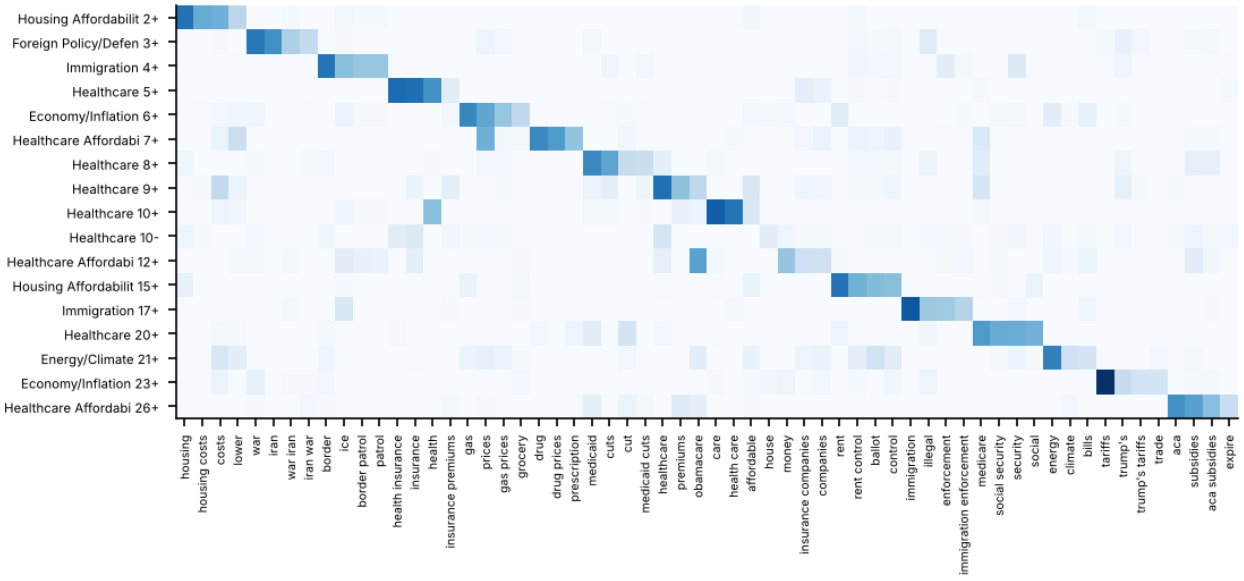


Figure 3. Term loadings U_k of the labelled poles (positive part)



2. Stage 1: sentiment and news → election odds

Table 2. Granger causality tests (topic sentiment → ΔProb)

Issue	Contract	Lags	F	p	q (BH)	Σγ	p (3d)	p (rev)
Economy/Inflation	House	1	0.685	0.409	0.741	-0.002	0.428	0.069
Economy/Inflation	Senate	1	0.050	0.823	0.983	-0.000	0.437	0.534
Economy/Inflation	Sweep	1	0.063	0.801	0.983	-0.001	0.407	0.154
Energy/Climate	House	1	0.006	0.940	0.983	-0.000	0.169	0.064
Energy/Climate	Senate	1	0.489	0.485	0.784	-0.002	0.202	0.352
Energy/Climate	Sweep	1	0.876	0.351	0.741	-0.002	0.334	0.242
Foreign Policy/Defense	House	1	3.802	0.053	0.557	0.008	0.265	0.989
Foreign Policy/Defense	Senate	1	2.496	0.116	0.625	0.008	0.957	0.765
Foreign Policy/Defense	Sweep	1	0.260	0.611	0.917	0.002	0.724	0.927
Healthcare	House	3	1.025	0.384	0.741	-0.001	0.762	0.481
Healthcare	Senate	1	2.234	0.137	0.625	-0.005	0.033	0.605
Healthcare	Sweep	3	2.832	0.041	0.557	-0.000	0.178	0.105
Healthcare Affordability	House	1	0.994	0.320	0.741	0.003	0.452	0.990
Healthcare Affordability	Senate	1	0.006	0.939	0.983	-0.000	0.883	0.233
Healthcare Affordability	Sweep	1	0.004	0.947	0.983	0.000	0.798	0.485
Housing Affordability	House	1	0.000	0.983	0.983	-0.000	0.772	0.716
Housing Affordability	Senate	1	1.508	0.221	0.741	-0.004	0.060	0.688
Housing Affordability	Sweep	1	0.763	0.384	0.741	-0.002	0.546	0.096
Immigration	House	1	2.107	0.149	0.625	0.002	0.157	0.664
Immigration	Senate	1	0.645	0.423	0.741	-0.003	0.443	0.308
Immigration	Sweep	1	0.158	0.691	0.968	-0.001	0.676	0.895

HC1 F-tests of lags of S_k in an AR model of ΔProb with lagged SPY, ΔVIX and Δ2Y controls; lags by BIC; q: BH FDR across the table; p (3d): 3-session pooled sentiment; p (rev): ΔProb → S_k.

Table 2B. GDELT news salience and tone → ΔProb (FDR q-values)

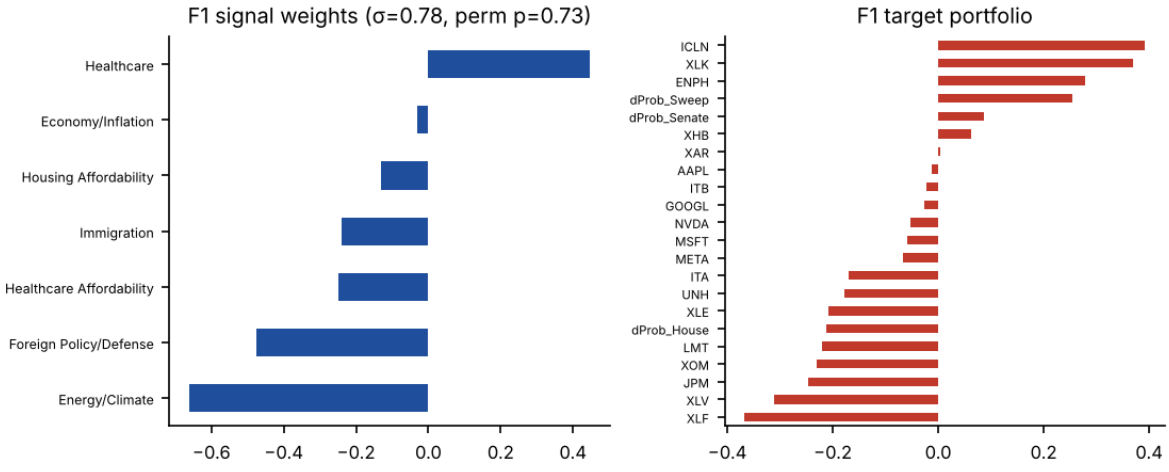
Issue	Sal House q	Sal Senate q	Sal Sweep q	Tone House q	Tone Senate q	Tone Sweep q	ρ X vs news
Economy/Inflation	0.89	0.72	0.72	0.94	1.00	0.86	0.31
Energy/Climate	0.86	0.89	0.49	0.89	0.89	0.72	-0.05
Foreign Policy/Defense	0.72	0.89	0.11	0.89	0.89	0.91	0.50
Healthcare	0.72	0.72	0.89	0.89	1.00	0.89	0.49
Healthcare Affordability	0.89	0.89	0.89	0.89	0.89	0.86	0.15
Housing Affordability	0.86	0.11	0.72	0.89	0.72	0.97	0.53
Immigration	0.14	0.96	0.97	0.89	0.89	0.96	0.07
Technology/AI	0.89	0.89	0.94	0.89	0.72	0.89	–

Same specification and FDR as Table 2; outage sessions drop out. ρ: correlation of X issue salience with GDELT's share of midterm coverage mentioning the issue (5-session means).

Covariance SVD. Signal–return alpha factors, trained to 2026-07-14, scored out of sample after it.

index	σ	perm p	oos_days	oos_corr	oos_t	sign_hit_rate
F1	0.783	0.735	62	-0.285	-2.306	0.403
F2	0.702	0.240	62	0.031	0.240	0.484
F3	0.484	0.539	62	-0.076	-0.588	0.452

Figure 6. Leading factor of the sentiment–return cross-covariance SVD



3. Stage 2: election odds → midterm basket

Table 3. Midterm equity basket reactivity matrix: next-session excess-return response Σγ (FDR q), ◆ = q < 0.10

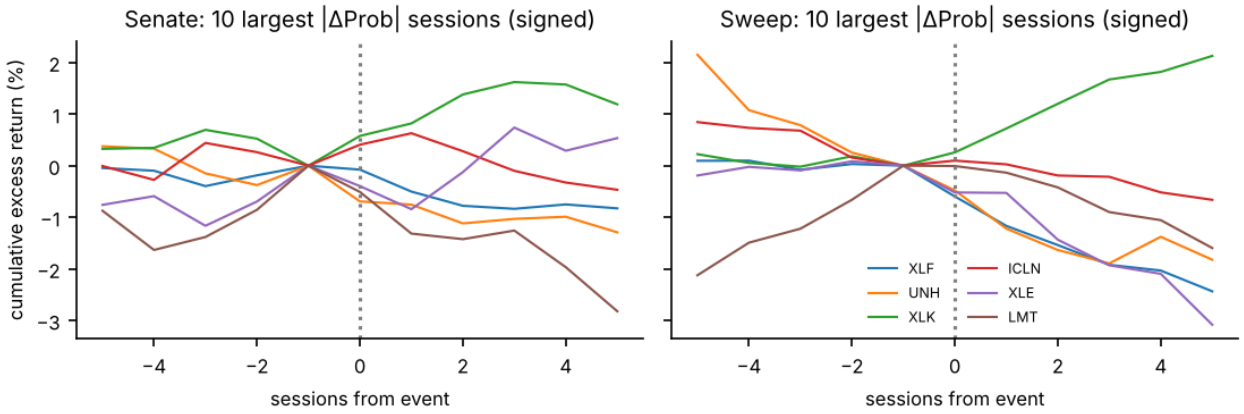
Asset	Ret: House	Ret: Senate	Ret: Sweep	Vol min q	β RS (Sweep) [95% CI]
XLE	-0.17 (0.73)	-0.28 (0.49)	-0.18 (0.67)	0.45	+0.09 [-5.6, 7.9]
ICLN	-0.48 (0.01) ◆	+0.03 (0.95)	+0.18 (0.49)	0.39	-0.15 [-8.3, 5.2]
XLV	+0.16 (0.51)	-0.02 (0.95)	-0.24 (0.11)	0.78	-0.38 [-4.3, 3.2]
XLF	-0.01 (0.97)	-0.10 (0.49)	-0.22 (0.05) ◆	0.45	+0.14 [-2.3, 3.0]
XAR	-0.11 (0.73)	+0.04 (0.95)	+0.01 (0.98)	0.56	-0.01 [-4.4, 4.9]
ITA	-0.08 (0.74)	-0.02 (0.95)	-0.10 (0.67)	0.67	+0.17 [-4.2, 5.0]
XLK	-0.11 (0.67)	+0.02 (0.95)	+0.29 (0.04) ◆	0.45	-0.17 [-4.7, 3.3]
ITB	+0.06 (0.95)	-0.01 (0.97)	-0.01 (0.98)	0.84	+0.30 [-3.4, 6.8]
XHB	-0.05 (0.95)	-0.00 (0.98)	-0.02 (0.95)	0.76	+0.06 [-2.7, 4.4]
ENPH	-0.69 (0.49)	-0.11 (0.95)	+0.59 (0.49)	0.67	-0.30 [-9.1, 8.9]
XOM	-0.22 (0.69)	-0.32 (0.49)	-0.20 (0.67)	0.45	-0.29 [-7.4, 6.9]
UNH	+0.34 (0.62)	-0.12 (0.73)	-0.46 (0.04) ◆	0.45	-0.40 [-3.7, 3.9]
JPM	-0.07 (0.95)	-0.13 (0.54)	-0.27 (0.19)	0.39	-0.21 [-4.0, 4.3]
LMT	-0.12 (0.80)	-0.11 (0.73)	-0.13 (0.73)	0.76	+0.70 [-5.5, 9.9]
NVDA	+0.00 (1.00)	+0.30 (0.06) ◆	+0.38 (0.06) ◆	0.48	-0.13 [-10.1, 7.0]
MSFT	+0.23 (0.54)	-0.16 (0.59)	-0.22 (0.49)	0.56	+1.46 [-11.7, 18.3]
GOOGL	+0.14 (0.92)	+0.11 (0.59)	-0.16 (0.62)	0.75	+0.75 [-10.6, 15.6]
META	+0.63 (0.27)	+0.53 (0.05) ◆	+0.41 (0.36)	0.39	-0.73 [-8.6, 7.2]
AAPL	-0.10 (0.75)	-0.10 (0.67)	+0.04 (0.95)	0.39	-0.07 [-6.3, 4.2]

Excess return over SPY; Vol: 2(H−L)/(H+L), smallest q across the three contracts; β RS: Rigobon–Sack response to ΔP(Sweep), H = top 20% GDELT news sessions. Full 57-row table with F, p and relevance tests: outputs/tables/table3_basket_reactivity.md.

Robustness battery for every FDR survivor (p-values)

cause	effect	sum_gamma	p	p_HAC	p_no_extremes	p_excl_dates	p_1st_half	p_2nd_half	p_placebo	robust
House	ICLN	-0.477	0.000	0.000	0.149	0.000	0.001	0.461	0.977	False
Sweep	XLF	-0.222	0.003	0.002	0.040	0.005	0.062	0.039	0.785	True
Sweep	XLK	0.289	0.002	0.002	0.006	0.003	0.129	0.003	0.190	True
Sweep	UNH	-0.456	0.002	0.002	0.004	0.003	0.015	0.064	0.465	True
Senate	NVDA	0.303	0.007	0.008	0.124	0.002	0.763	0.002	0.290	False
Sweep	NVDA	0.379	0.006	0.006	0.051	0.009	0.405	0.003	0.185	True
Senate	META	0.534	0.004	0.003	0.051	0.003	0.383	0.019	0.952	True

Figure 7. Basket reactions around large prediction-market moves



4. Methods

Data and timing. All series sit on the NYSE calendar. Tweets are fetched per session (previous 16:00 ET close to this one); prediction-market probabilities are the last hourly quote at or before 16:00 ET (Kalshi CONTROLH-2026, CONTROLS-2026, KXBALANCEPOWERCOMBO-27FEB; Polymarket control and balance-of-power markets; each normalised across its outcomes, consensus = equal-weight average, PredictIt a zero-weight cross-check); a GDELT day counts for the next session. Text: twitterapi.io, a general issue × political-context stratum plus an affordability oversample used for tone only; English, no retweets/replies, 5–150 likes, followers ≥ 10, views ≥ 100, astroturf removed (93,450 credits, every call ledgered).

LSA and sentiment. Uncentred TF-IDF (uni/bigrams, hashtags, cashtags) → truncated SVD, k = 30 → varimax rotation of components 1..29 (centring would densify the matrix and treat unseen terms as negative evidence). Poles are labelled by the share of loading energy on an issue's seed terms (≥ 15%, purity ≥ 60%). A document's issue weight is the squared cosine of its projection on that issue's poles; polarity is VADER with an issue-specific overlay ('rising' is negative for prices and rents). $S_{k,t} = \Sigma(\text{Pos} - \text{Neg})/\Sigma\text{Total}$ pooled over 5 sessions with ≥ 5 effective documents.

Econometrics. ADF-tested series; Granger F-tests (HC1, BIC lags, lagged SPY/ΔVIX/Δ2Y controls); Benjamini–Hochberg FDR per table; a robustness battery for every survivor (HAC, without the 5 largest ΔProb sessions, without 5 Jun 2026, both halves, placebo). Covariance SVD of $(1/T)\Sigma\, s_{t-1}'\delta y_t$ on the first 60% of the sample with a circular-shift permutation test, scored out of sample on the last 40%. Rigobon–Sack $\beta = \Delta\text{Cov}/\Delta\text{Var}$ across H (top 20% GDELT midterm-news sessions) and L sessions, outages excluded, regime bootstrap.

5. Limitations

• One campaign (155 sessions): daily tests have little power against effects that build over weeks, and the robust Stage 2 results are hypotheses for 2028, not a trading rule.

• The X sample is politically engaged mid-size accounts (median ~24–42k followers), not the median voter; Technology/AI never formed a topic, so Big Tech is tested only against odds.

• GDELT had 19 outage days (mid-September, most of October), excluded rather than read as quiet. Yahoo closes for SPY/IVV/VOO disagree day by day over 25 Jun – 1 Jul 2026; a median benchmark or dropping that week changes no conclusion. Full checks: DATA_AUDIT.md.

AI use: built with Claude Code; see AI_USE.md for what was done by the model and by the author.